

Scientific Machine Learning: Blending Science with Data

Homework Day 8

You are given the function

$$f(x; b) = x^3 \exp(-(x/b)^2) - 1.0$$

Consider the values of $b \in [1.5, 4.0]$. Your values of x must satisfy $f(x; b) = 0$

- a) Plot f vs x for different values of b to understand how the values of x that satisfy $f(x, b) = 0$ changes with b .
- b) **(LEVEL 1)** Write a program in Julia that finds the value of b for $x = 10$. Use your result in a) to get your initial guess(es).
- c) **(LEVEL 3)** Write a Julia program to check your answer in part b).